Supplementary Material: Conchordal: Emergent Harmony via Direct Cognitive Coupling in a Psychoacoustic Landscape

Koichi Takahashi[1,2]
[1]Institute for Advanced Biosciences, Graduate School of Media and Governance, Keio University
[2]Conchordal.org
info@conchordal.org

S1. Algorithm Pseudocode

S1.1 Pitch Adaptation (Experiment 1)

Input: agents $\{a_1, \ldots, a_N\}$ with pitches x_i in $\log_2$-freq space.
Parameters: step size $\lambda = 15\,\text{ct}$, crowding weight $\lambda_C = 0.15$, $T_0 = 0.05$, $\tau = 30$.

`for` round $t = 1$ `to` 100:
  $T_t \leftarrow T_0 \cdot \exp(-t/\tau)$
  `for` each agent a_i:
    $\delta \sim \text{Uniform}(-\lambda, +\lambda)$; $x' \leftarrow x_i + \delta$
    $P(x) = \lambda_C \sum_{j \neq i} K_{\text{crowd}}(\Delta_{\text{erb}}(x, x_j))$
    $S_{\text{old}} \leftarrow s(t) \cdot C_{\text{field}}(x_i) - P(x_i)$
    $S_{\text{new}} \leftarrow s(t) \cdot C_{\text{field}}(x') - P(x')$
    `if` $U(0,1) < \min\big(1, \exp\big((S_{\text{new}} - S_{\text{old}})/T_t\big)\big)$ `then` $x_i \leftarrow x'$

where $s(t) = -1$ for $t \leq 50$ (dissonance-seeking), $+1$ for $t > 50$ (consonance-seeking).

S1.2 Metabolic Policy (Experiment 2)

Parameters: $c_b = 0.5$/s, $c_a = 0.02$, $r_E = 0.5$, $\Delta t = 512/48000\,\text{s}$.

`for` each time step:
  `for` each agent a_i:
    $E_i \leftarrow E_i - c_b \cdot \Delta t$ *(basal drain)*
    `if` phase ϕ_i crosses zero `and` $E_i > 0$: *(attack event)*
      $C_{\text{level01}} \leftarrow \sigma\big(2 \cdot C_{\text{field}}(x_i)\big)$
      $E_i \leftarrow E_i - c_a + r_E \cdot C_{\text{level01}}$
    `if` $E_i \leq 0$: agent dies; respawn at random pitch within ± 1 oct

S1.3 Kuramoto Phase Coupling (Experiment 3)

Parameters: $k_{\text{time}} = 3.0$, $\lambda_v = 1.0$, $v_{\text{floor}} = 0.0$, $E_{\text{cap}} = 1.0$, kick at 2 Hz.

`for` each time step dt:
  `for` each agent a_i:
    $v_i \leftarrow (E_i/E_{\text{cap}})^{0.5}$ *(vitality)*
    $g_i \leftarrow \text{clamp}\big((v_i - v_{\text{floor}})/(1 - v_{\text{floor}}),\ 0,\ 1\big)$
    $k_{\text{eff},i} \leftarrow k_{\text{time}} \cdot \lambda_v \cdot g_i$
    $\dot{\phi}_i \leftarrow \omega_i + (k_{\text{eff},i}/N) \sum_j \sin(\phi_j - \phi_i) + F(t)$
    $\phi_i \leftarrow \phi_i + \dot{\phi}_i \cdot dt$
  $R(t) = (1/N)\big|\sum_j e^{i\phi_j}\big|$ *(order parameter)*

Continuous energy model: $dE_i/dt = -c_b + r_E \cdot C_i$ ($c_b = 0.05$/s, $r_E = 0.1$/s).

S1.4 Hereditary Respawn (Experiment 4)

Parameters: mutation $\sigma = 0.003\ \log_2$, crowding weight λ_C.

`for` each time step:
  Run metabolic policy (S1.2) with recharge enabled
  `if` agent a_i dies ($E_i \leq 0$):
    Select parent a_p uniformly from surviving agents
    $x_{\text{new}} \sim \mathcal{N}(x_p, \sigma^2)$; clamp to anchor ± 1 oct
    Respawn at x_{new} with $E = 1.0$

Random control: respawn at $\text{Uniform}(\text{anchor} \pm 1\,\text{oct})$ instead.

S2. Normalization of H_{01} and R_{01}

Both harmonicity and roughness are normalized to $[0, 1]$ before entering the consonance core.

Harmonicity (H_{01})

The raw harmonicity potential from sibling projection (Eqs. 1–2 in the main text) is normalized by dividing by the self-harmonicity of a single harmonic tone at the reference position (which equals the kernel's maximum response):

$$H_{01}(x) = \text{clamp}\left(\frac{H_{\text{raw}}(x)}{H_{\text{ref,max}}},\ 0,\ 1\right), \tag{S1}$$

where $H_{\text{ref,max}} = 1.0$ (the kernel is pre-normalized so that a single harmonic stack produces unit peak).

Roughness (R_{01})

Raw roughness is first expressed as a ratio $\rho = R_{\text{raw}}/R_{\text{ref}}$, where R_{ref} is the roughness of a single-tone reference density (computed once at initialization). This ratio is then

mapped through a piecewise saturation function with physiological constant k ($k = 1.0$ in all experiments):

$$R_{01}(\rho) = \begin{cases} 0 & \rho \le 0, \\ \dfrac{\rho}{1+k} & 0 < \rho < 1, \\ 1 - \dfrac{k}{\rho + k} & \rho \ge 1, \end{cases} \tag{S2}$$

clamped to $[0, 1]$. For $\rho < 1$ (below-reference roughness) the mapping is linear; for $\rho \ge 1$ it is a hyperbolic saturation that approaches 1 asymptotically as $\rho \to \infty$. At $\rho = 1$ the function is continuous: $R_{01}(1) = 1/(1 + k)$. With $k = 1$, the reference roughness maps to $R_{01} = 0.5$.

S3. Complete Parameter Table

Table S1 lists all parameters used across experiments.

S4. Experiment 4: Full Sweep Results

Table S2 reports the hereditary adaptation parameter sweep. Each of the 15 conditions was run with 20 seeds per arm (heredity vs. random control). The reported setting in the main text ($\sigma = 0.003$, deaths $= 4{,}000$, $N = 32$) is marked with $\star$.

S5. Bilinear Consonance Core: Derivation

The consonance core (Eq. 4 in the main text) is the most general bilinear form for two inputs $H_{01}, R_{01} \in [0, 1]$:

$$C = a\, H_{01} + b\, R_{01} + c\, H_{01}\, R_{01} + d. \tag{S3}$$

$C_{\textbf{field}}$ $(a, b, c, d) = (1, -1.35, 1, 0)$:

$$C_{\text{field}} = H_{01} - 1.35\, R_{01} + H_{01}\, R_{01} \tag{S4}$$
$$= H_{01}(1 + R_{01}) - 1.35\, R_{01}. \tag{S5}$$

When H_{01} is high (strong harmonicity), the $H_{01}\, R_{01}$ term partially compensates for roughness, reflecting the perceptual observation that harmonically rich intervals tolerate moderate beating. When $H_{01} = 0$, roughness is penalized at full strength.

$C_{\textbf{density}}$ $(a, b, c, d) = (1, 0, -1, 0)$:

$$C_{\text{density}} = H_{01}(1 - R_{01}). \tag{S6}$$

This product is nonnegative and peaks where harmonicity is high and roughness is low. In Conchordal's interactive mode it serves as a spawn-position weight; it is not used in the experiments reported in this paper but is shown in Figure 1D for comparison.

$C_{\textbf{level01}}$: C_{field} passed through a logistic sigmoid $\sigma(\beta(x - \theta))$ with $\beta = 2, \theta = 0$. This bounds the metabolic signal to $[0, 1]$ while preserving the ranking.

S6. Sign-Flip Permutation Test

The interaction contrast for the integration test (Figure 7 in the main text) is tested with a one-sample sign-flip permutation test:

1. For each of $n = 20$ matched seeds, compute the contrast $d_i = \mathrm{HH}_i - \mathrm{H}_i - \mathrm{RH}_i + \mathrm{R}_i$.
2. Under H_0: $E[d_i] = 0$, each d_i is equally likely to be positive or negative. For each of $2^{20} = 1{,}048{,}576$ sign patterns $s \in \{-1, +1\}^{20}$, compute $\bar{d}_s = \frac{1}{n} \sum_i s_i\, d_i$.
3. The p-value is the fraction of sign patterns yielding $|\bar{d}_s| \ge |\bar{d}_{\text{obs}}|$.

Because $n = 20$, exact enumeration is feasible. The observed $p = 0.000053$ (1 in 18,868 patterns) with mean contrast 0.30 (95% CI $[0.20, 0.39]$).

S7. Consonance-Only Control (Experiment 1)

Experiment 1 uses a two-phase curriculum: dissonance-seeking (steps 0–24) followed by consonance-seeking (steps 25–49). To assess whether the resulting polyphony depends on this schedule, we ran a *consonance-only* control with identical parameters but $s(t) = +1$ for all 50 sweeps (*local-search* condition, 20 seeds).

Both conditions reach indistinguishable final consonance ($C_{\text{score}} = 0.731$; $p = 0.98$), confirming that consonant structure emerges from the landscape–penalty interaction regardless of schedule. However, the curriculum significantly improves pitch diversity: more unique bins (9.4 vs. 5.9; $p < 0.0001$) and wider nearest-neighbour spacing (0.324 vs. 0.049 st; $p = 0.0001$), while interval entropy does not differ significantly ($p = 0.19$). The dissonance-seeking phase thus serves as an initialisation strategy that broadens the pitch distribution, preventing premature collapse to a small number of positions.

S8. Terrain Control and Robustness Analysis

Three supplementary analyses test whether the self-organised polyphony reported in Experiment 1 depends on the specific structure of the psychoacoustic landscape rather than on incidental parameter choices.

S8.1 Shuffled-Landscape Control

We destroy spatial coherence by applying a fixed random permutation of the C_{score} bin indices, generated once per run and held fixed throughout all sweeps. Agents therefore move on a stationary landscape with the same *marginal* score distribution but no local gradient structure. Table S4 compares the shuffled condition with the standard *local-search* condition ($n = 20$ seeds each).

C_{score} (ref) denotes the consonance score of each final configuration evaluated on the *original* (unshuffled) landscape, enabling a fair cross-condition comparison. Without

coherent spatial gradients, agents still spread across the register (similar unique-bin count and nearest-neighbour spacing) thanks to the crowding penalty, but the interval structure degrades sharply: entropy rises from 1.52 to 3.36 nats ($p < 0.0001$), and C_{score} drops from 0.731 to 0.459 ($p < 0.0001$). The coherent psychoacoustic landscape is therefore necessary for agents to self-organise into musically structured pitch configurations.

S8.2 Harmonicity–Roughness Misalignment Control

The consonance kernel combines harmonicity (H_{01}) and roughness (R_{01}). We test whether their spatial alignment matters by circularly shifting R_{01} by Δ cents before scoring, which preserves each component's marginal statistics but breaks the psychoacoustically motivated registration between them. Four shift values span the range $\Delta \in \{137, 271, 389, 523\}$ cents. Table S5 reports the results ($n = 20$ seeds per shift, 80 runs total for the misaligned aggregate).

All four misaligned conditions produce significantly fewer unique pitch bins than the local-search baseline (all $p < 10^{-4}$, Welch t), dropping from 9.4 to $\approx$6 bins—comparable to the consonance-only control (S7). Nearest-neighbour distances also decrease significantly (all $p \leq 0.0002$). Entropy effects are mixed: two shifts reach $p < 0.05$ while two do not, and the aggregate is significant ($p = 0.0002$). Misalignment thus preserves consonance (all $C_{\text{score}} \geq 0.70$) but eliminates the diversity benefit of the curriculum, reinforcing the conclusion from the shuffled-landscape control (S8.1) that a coherent landscape gradient is essential for structured polyphony.

S8.3 Consonance Kernel Coefficient Sensitivity

The consonance kernel $C = a\,H_{01} + b\,R_{01} + c\,H_{01}\,R_{01} + d$ has four coefficients; the default is $(a, b, c, d) = (1.0, -1.35, 1.0, 0.0)$. We sweep $b \in \{-2.0, -1.35, -0.7, 0.0\}$ and $c \in \{0.0, 0.5, 1.0, 1.5\}$ with $a = 1$, $d = 0$ fixed (16 conditions, 20 seeds each). Table S6 reports the results; because the sweep varies the kernel while holding everything else constant, it isolates the sensitivity of the consonance landscape to its parameterisation.

Unique-bin counts (5.8–6.5) and entropy (1.61–1.84 nats) are stable across the entire (b, c) grid, showing that the emergent pitch structure is not fragile to kernel parameterisation. C_{score} increases when the roughness penalty $|b|$ decreases or the interaction weight c increases (0.729–0.814), which is expected from the kernel definition. The overall level of unique bins ($\approx$6) matches the consonance-only control (S7), confirming that the crowding penalty—not the kernel coefficients—is the primary driver of the higher diversity (9.4 bins) seen in the full local-search condition.

S8.4 Integration Test Terrain Control (Experiment 4)

To test whether the synergistic heredity$\times$hill-climbing interaction (main text, §5.1) depends on the coherent structure of the psychoacoustic landscape, we repeated all four integration-test conditions on a shuffled landscape, yielding a complete $2 \times 2 \times 2$ factorial (heredity/random $\times$ hill-climbing on/off $\times$ real/shuffled terrain; 8 conditions $\times$ 20 seeds $= 160$ runs).

On real terrain, the interaction contrast is large and significant (0.30, $p < 0.001$); on the shuffled landscape the contrast drops to 0.03 ($p = 0.19$), eliminating the synergy entirely. The full factorial also reveals that coherent terrain affects the non-hill conditions asymmetrically: heredity benefits substantially from real terrain even without hill-climbing (H real 0.185 vs. H shuffled 0.051, $p = 0.0002$), because consonance-gated metabolism rewards agents that happen to inherit consonant positions, an effect that vanishes when the landscape is spatially incoherent. Random respawn shows no such terrain dependence (R real 0.061 vs. R shuffled 0.059, $p = 0.70$), confirming that the metabolism-mediated terrain effect operates specifically through heritable pitch structure.

S8.5 Topology Generalization (Experiment 4)

To test whether the heredity$\times$hill-climbing interaction depends on the specific harmonicity profile used in the default configuration, we repeated the four real-terrain conditions (H, HH, R, RH; 20 seeds each) under three landscape topologies, generated by varying the number of environment partials in the anchor harmonic series (3, 6, 10). Fewer partials produce a smoother, broader harmonicity landscape; more partials increase spectral fine structure.

The interaction contrast is large and significant ($p < 0.001$) across all three topologies, ranging from 0.30 to 0.37, confirming that the synergistic effect is not specific to the default harmonicity profile but generalizes across qualitatively different landscape structures.

S8.6 Phase Coherence (Experiment 4)

Each Experiment 4 agent entrains to a shared theta oscillator via a Kuramoto coupling mechanism. We computed the population Kuramoto order parameter $R = |N^{-1} \sum_j e^{i(\phi_j - \phi_\theta)}|$ at the final snapshot, where ϕ_j is each agent's rhythm phase and ϕ_θ is the global theta phase ($R = 1$ indicates perfect synchrony; $R \approx 0$ indicates incoherence).

Phase coherence is low across all conditions ($R \approx 0.12$–0.18) and does not differ significantly between heredity and random conditions ($p > 0.4$). This indicates that in the current Experiment 4 parameterization, consonance-gated entrainment does not produce differential temporal binding across conditions within 4,000 deaths. On real terrain, heredity-type conditions show slightly higher R than

their shuffled counterparts (e.g., H: 0.183 vs. 0.117), consistent with the consonance-gated metabolic advantage providing agents more stable positions from which to entrain; random conditions show no such terrain dependence (R: 0.169 vs. 0.165).

Table S1: Complete parameter listing for all experiments.

Parameter	**Symbol**	**Exp. 1**	**Exp. 2**	**Exp. 3**	**Exp. 4**
Anchor frequency	f_0	220 Hz	220 Hz	220 Hz	220 Hz
Population size	N	24	32	32	32
Seeds		20	20	20	20
Pitch range		± 1 oct	± 1 oct	± 1 oct	± 1 oct
Harmonicity / Roughness					
Harmonic order	N_H	16	16	16	16
Power-law exponent	ρ	0.4	0.4	0.4	0.4
Roughness saturation	k	1.0	1.0	1.0	1.0
Consonance core					
C_{field} coefficients	(a, b, c, d)	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$	$(1, -1.35, 1, 0)$
C_{density} coefficients[‡]	(a, b, c, d)	$(1, 0, -1, 0)$	—	—	—
Sigmoid (β, θ)		—	$(2, 0)$	—	$(2, 0)$
Pitch adaptation					
Step size	λ	0.15 st	—	—	—
Crowding weight	λ_C	0.15	—	—	0.15
Initial temperature	T_0	0.05	—	—	—
Annealing time constant	τ	30 steps	—	—	—
Duration		50 sweeps	—	—	—
Phase switch		step 25	—	—	—
Burn-in discarded		10 steps	—	—	—
Metabolism					
Basal cost	c_b	—	0.5/s	0.05/s[†]	0.5/s
Attack cost	c_a	—	0.02	—	0.02
Recharge coefficient	r_E	—	0.5/att	0.1/s[†]	0.5/att
Energy cap	E_{cap}	—	1.0	1.0	1.0
Time step	Δt	—	512/48k s	0.02 s	512/48k s
Termination		—	200 deaths	40 s	4000 deaths
Temporal coupling					
Base coupling	k_{time}	—	—	3.0	—
Vitality sensitivity	λ_v	—	—	1.0	—
Vitality floor	v_{floor}	—	—	0.0	—
External kick	f_{kick}	—	—	2 Hz	—
Intrinsic frequency	ω_0	—	—	$1.8 \cdot 2\pi$	—
Frequency jitter		—	—	$\pm 2\%$	—
Heredity					
Mutation σ	σ_m	—	—	—	$0.003\ \log_2$
Parent selection		—	—	—	uniform
JI cross-validation					
Proximity kernel	α	—	—	—	4.0×10^{-4} ct^{-2}
Ratio limit	p, q	—	—	—	≤ 8
Ratio weight		—	—	—	$1/(pq)$

[†] Experiment 3 uses a simplified continuous energy model (see S1.3).
[‡] Visualization only (Figure 1D); not used in experiment algorithms.

Table S2: Full parameter sweep for Experiment 4 (hereditary adaptation). Each row reports one parameter setting; "Sep." is the mean ΔC_{score}(heredity) $-$ mean ΔC_{score}(random). Welch's t-test compares the two arms ($n = 20$ seeds each). $\star$ marks the setting reported in the main text.

σ	Deaths	N	$\overline{\Delta C}_{\text{her}}$	$\overline{\Delta C}_{\text{rnd}}$	Sep.	t	p
0.001	2000	32	0.074 ± 0.150	0.004 ± 0.035	0.071	2.06	0.052
0.001	3000	32	0.117 ± 0.174	0.007 ± 0.025	0.111	2.81	0.011*
0.002	2000	32	0.067 ± 0.119	0.004 ± 0.035	0.063	2.29	0.032*
0.002	3000	32	0.039 ± 0.093	0.007 ± 0.025	0.032	1.49	0.151
0.003	2000	16	0.035 ± 0.101	-0.004 ± 0.036	0.039	1.62	0.118
0.003	2000	24	0.089 ± 0.155	0.004 ± 0.029	0.085	2.41	0.025*
0.003	2000	32	0.096 ± 0.138	0.004 ± 0.035	0.093	2.91	0.008*
0.003	2000	48	0.061 ± 0.141	0.004 ± 0.021	0.058	1.80	0.087
0.003	3000	32	0.118 ± 0.160	0.007 ± 0.025	0.111	3.07	0.006*
$\star$0.003	4000	32	0.121 ± 0.177	-0.003 ± 0.027	0.124	3.10	0.006*
0.003	5000	32	0.113 ± 0.167	-0.003 ± 0.025	0.116	3.07	0.006*
0.005	2000	32	0.078 ± 0.135	0.004 ± 0.035	0.075	2.40	0.026*
0.005	3000	32	0.072 ± 0.119	0.007 ± 0.025	0.065	2.39	0.027*
0.01	2000	32	0.058 ± 0.102	0.004 ± 0.035	0.054	2.25	0.034*
0.02	2000	32	0.106 ± 0.135	0.004 ± 0.035	0.102	3.26	0.004*

*Significant at $p < 0.05$. $\star$ = reported setting in the main text ($\sigma = 0.003$, 4,000 deaths, $N = 32$).

Table S3: Curriculum vs. consonance-only control (Experiment 1, *local-search* condition, $n = 20$ seeds).

Metric	**Curriculum**	**Cons.-only**	t
C_{score} (post)	0.731 ± 0.057	0.731 ± 0.041	-0.02
Unique pitch bins	9.4 ± 1.2	5.9 ± 0.6	-11.6^{***}
NN distance (st)	0.324 ± 0.245	0.049 ± 0.068	-4.8^{***}
Interval entropy	1.52 ± 0.36	1.65 ± 0.27	1.4

Table S4: *Local-search* vs. shuffled landscape (Experiment 1).

Metric	**Local-search**	**Shuffled**	t
Unique pitch bins	9.4 ± 1.2	9.6 ± 1.0	0.6
NN distance (st)	0.324 ± 0.245	0.328 ± 0.207	0.1
Interval entropy	1.52 ± 0.36	3.36 ± 0.27	18.6^{***}
C_{score} (ref)	0.731 ± 0.057	0.459 ± 0.042	-17.3^{***}

Table S5: *Local-search* vs. H/R-misaligned conditions (Experiment 1).

Condition	Unique bins	NN dist. (st)	Entropy
Local-search	9.4 ± 1.2	0.324 ± 0.245	1.52 ± 0.36
$\Delta = 137\,¢$	5.9 ± 1.0	0.047 ± 0.081	1.72 ± 0.30
$\Delta = 271\,¢$	6.4 ± 0.8	0.062 ± 0.078	1.83 ± 0.29
$\Delta = 389\,¢$	6.2 ± 1.0	0.065 ± 0.103	1.74 ± 0.27
$\Delta = 523\,¢$	5.8 ± 0.9	0.032 ± 0.054	1.57 ± 0.31
Misaligned (avg)	6.1 ± 1.0	0.052 ± 0.082	1.72 ± 0.31

Table S6: Coefficient sweep: polyphony metrics across 16 (b, c) settings.

b	c	Unique bins	Entropy	C_{score}
-2.00	0.00	6.2 ± 1.0	1.65 ± 0.26	0.738 ± 0.053
-2.00	0.50	5.8 ± 1.1	1.61 ± 0.33	0.729 ± 0.051
-2.00	1.00	6.2 ± 1.1	1.63 ± 0.26	0.746 ± 0.043
-2.00	1.50	6.1 ± 1.2	1.67 ± 0.29	0.751 ± 0.042
-1.35	0.00	6.2 ± 1.2	1.66 ± 0.28	0.736 ± 0.047
-1.35	0.50	6.2 ± 1.0	1.67 ± 0.32	0.733 ± 0.048
-1.35	1.00	6.4 ± 0.7	1.84 ± 0.29	0.734 ± 0.035
-1.35	1.50	5.9 ± 1.1	1.61 ± 0.33	0.760 ± 0.043
-0.70	0.00	6.5 ± 1.0	1.74 ± 0.28	0.739 ± 0.042
-0.70	0.50	6.1 ± 1.0	1.71 ± 0.31	0.761 ± 0.034
-0.70	1.00	5.8 ± 1.2	1.61 ± 0.33	0.768 ± 0.040
-0.70	1.50	6.2 ± 0.8	1.66 ± 0.22	0.788 ± 0.041
0.00	0.00	6.3 ± 1.2	1.68 ± 0.31	0.763 ± 0.034
0.00	0.50	5.9 ± 1.1	1.75 ± 0.28	0.760 ± 0.042
0.00	1.00	6.2 ± 0.8	1.79 ± 0.27	0.792 ± 0.034
0.00	1.50	5.8 ± 0.9	1.71 ± 0.23	0.814 ± 0.028

Table S7: Full $2\times2\times2$ integration terrain control (20 seeds each).

Condition	Real	Shuffled	Diff.
H (her., no hill)	$.185 \pm .076$	$.051 \pm .006$	$.134^{***}$
HH (her. + hill)	$.530 \pm .068$	$.082 \pm .049$	$.449^{***}$
R (rnd., no hill)	$.061 \pm .008$	$.059 \pm .008$	.003
RH (rnd. + hill)	$.107 \pm .013$	$.056 \pm .006$	$.051^{***}$

Interaction contrast (real): 0.30 [0.20, 0.39], $p < 0.001$
Interaction contrast (shuffled): 0.03 [-0.02, 0.08], $p = 0.19$

Table S8: Topology generalization: interaction contrast across different numbers of environment partials (20 seeds each).

Partials	H	HH	R	RH	Contrast [95% CI]
3	.152	.563	.074	.116	$.37\ [.29, .45]^{***}$
6 (default)	.185	.530	.061	.107	$.30\ [.21, .39]^{***}$
10	.125	.528	.064	.108	$.36\ [.28, .44]^{***}$

$^{***}p < 0.001$.

Table S9: Phase coherence R (Kuramoto order parameter) at the final snapshot (20 seeds each).

Condition	Real terrain	Shuffled
H (heredity, no hill)	0.183 ± 0.027	0.117 ± 0.027
HH (heredity + hill)	0.163 ± 0.036	0.125 ± 0.033
R (random, no hill)	0.169 ± 0.032	0.165 ± 0.029
RH (random + hill)	0.183 ± 0.041	0.181 ± 0.039

HH $-$ RH: -0.020 [-0.078, 0.039], $p = 0.53$
H $-$ R: 0.014 [-0.022, 0.051], $p = 0.47$